



ASHHAR AHMED

SENIOR SOFTWARE ENGINEER

A/A

CONTACT

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- LinkedIn
- GitHub
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- Blog Portfolio

TECH SKILLS

- Elixir 80%
- Node/Express js 70%
- React 18/19 TS 70%
- Angular 7/18 65%
- Python3 FastAPI 70%
- Google ADK / Gemini Live 65%
- LangGraph 70%
- MCP 60%
- LLM Integration 60%
- Docker, K8, Helm, Jenkins, Git 65%
- Mysql, PG, Redis, pgVector 65%
- Agile /Scrum 70%
- Google ES/CX, Wireshark 60%
- Linux (CentOS) 65%
- Backstage, Mkdocs 65%

LANGUAGES

- English (Fluent)
- Tamil (Fluent)
- Sinhala (Basic)



PROFILE

Results-driven Senior Software Engineer with over 5.5 years of experience in enterprise solution development, system architecture, and team leadership across the telecommunications, PaaS, and SaaS domains. Proven track record at Global Wavenet and Virtusa in delivering innovative, high-performance solutions and managing complex projects. Currently leading the architecture and development of an AI Agent Orchestration platform - building multi-agent, multi-channel AI workflows using Python/FastAPI, Google ADK, LangGraph, and MCP. Passionate about continuous learning, personal development, and contributing to impactful software products.



WORK EXPERIENCE

Senior Software Engineer

Global Wavenet | Jul 2024 - Present

- Architected and led development of the Compose AI Agent Orchestrator - a Python/FastAPI service enabling no-code configuration and execution of multi-agent AI flows across IVR, WhatsApp, Telegram, and web channels.
- Designed an 8-layer system including GraphExecutor, BundleLoader, AgentFactory, GuardrailRunner, and StateManager.
- Integrated Google ADK with Gemini Live native audio for real-time voice IVR and built a LangGraph adapter for model-agnostic digital channels (GPT-4o, Claude, Gemini).
- Implemented MCP (Model Context Protocol) tool integration for AI agents, supporting both platform-level and per-tenant customer tool bindings.
- Built a multi-tier AI guardrail system: rules/regex (Tier 1), ML toxicity detection via Detoxify (Tier 2), and LLM-as-Judge hallucination checking (Tier 3).
- Implemented session state management with Redis and PostgreSQL with pgvector for RAG/vector-search capabilities; observability via OpenTelemetry, Grafana, and Jaeger.
- Upgraded the entire Compose application frontend from Angular 7 to React 19 TypeScript (xyflow canvas with 37+ node types).
- Contributed to low-code API development and integration with WSO2 API Manager; actively resolved Level 3 production issues and participated in feature design and planning.
- Played a key role in developing AI-based flow execution within the platform.

Software Engineer

Global Wavenet | Apr 2022 - Jul 2024

- Developed and deployed middleware solutions for major telecom clients including Tigo Panama, Tigo Colombia, Dialog, and Hutch.
- Selected to join the Research & Development team to build a core omnichannel orchestration PaaS platform as a full-stack engineer.
- Designed and developed modular API endpoints, bundled them as microservices, and deployed them within a cloud-native architecture.
- Explored and customized Backstage plugins (polygot development) to enhance internal developer experience and visibility.
- Temporarily contributed to a multivendor marketplace project, focusing on architecture review and initial development analysis.

Associate Software Engineer

Global Wavenet | Feb 2021 - Apr 2022

- Developed IVR, USSD, SMS, and ESB middleware using Node.js for telecom clients.
- Led end-to-end delivery of 249 IVR call flow application for Claro Colombia, from requirements to deployment.
- Integrated Google Dialog flow ES/CX for voice recognition in IVR workflows.
- Managed the full project lifecycle, ensuring timely and high-quality delivery.

Software Engineer Intern

Virtusa | Jul 2018 - Aug 2019

- Contributed to the Pigeon incident management system for the Colt ASM project.
- Worked with Angular, Spring Boot, MySQL, and RabbitMQ in a CentOS environment.
- Focused on automating incident creation through email parsing and mailbox integration.

CERTIFICATIONS

- Agile scrum certification
- WSO2 APIM and IS Trainning
- Certificate in JAVA
- NDG Linux
- Advanced JavaScript
- Introduction to DevOps
- Azure DevOps
- Containerize SpringBoot and Express App & Deploy on Azure

OTHER SKILLS

- Teamwork
- Willing to learn & quick Learner
- Solution Design
- Ability work under pressure
- Self Motivation
- Time Management
- Leadership
- Effective Communication
- Critical Thinking

REFERENCE

J M B K B Thilakaratne,
B.Sc (Hons) Computing (upper second class)
Oxford Brookes University
Headington Campus
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Associate Software Architect
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EDUCATION

BSc (Hons)in IT, Specialized in Software Engineer

Sri Lanka Institute of Information Technology 2017 -Feb - 2021-Feb

- CGPA: 3.2
- Majored in Software Engineering
- 2nd , 3rd , and 4th year Batch representative 2018, 20019 and 2020
- Editor of Majlis SLIIT
- Thesis: FUI Generator Tool

G.C.E Advanced Level Biological Science

A/ Zahira National School Jan.2012 - Sep.2015

- Chemistry - C, Physics - C, Biology - C, Z-score = 0.97

G.C.E(O/L)- 2011

MR/ Zahira National School Jan.2001 - Dec.2011

- 5A, 2B, 2C with (Mathematics C, English B Science A & ICT A)5A, 2B, 2C with (Mathematics C, English B Science A & ICT A)



PROJECTS

Sayura (Compose) - Omnichannel Orchestration PaaS Platform

Technologies: Elixir/Phoenix, Node.js, React 19 (xyflow), Python/FastAPI, Google ADK, LangGraph, Gemini Live, LLM (Gemini · Claude · GPT-4o), Redis, PostgreSQL/pgvector, NATS, OpenTelemetry, MySQL, Kubernetes

- Ongoing R&D at Global Wavenet. Evolved from a telecom omnichannel PaaS into a full AI Agent Orchestration Platform with an 8-layer architecture
- Ongoing development as part of the R&D team at Global Wavenet.
- Core focus on enabling low-code/no-code creation of APIs, microservices, and legacy and digital channel integration (IVR, USSD, SMS, WhatsApp, Telegram, Webchat, Messenger, Viber).
- Delivered solutions to telecom clients including Safaricom, Ncell, Robi, Claro Colombia, Tigo (Colombia, Bolivia, Honduras), Ordio, Mobital, Dialog, Hutch, SLT, and others.

Hypermart - Multi-vendor Marketplace Platform

Technologies: Angular 18, Express (TypeScript), gRPC, RabbitMQ, REST APIs

- Contributed to platform architecture and backend development using microservices.
- Worked with a team over a 5-month period to implement Node.js services and train developers.
- Served enterprise clients such as Hutch, Encell, and SLT.

Pigeon - Incident Management System (Colt ASM Project)

Technologies: Angular 2/7, Spring Boot, MySQL, RabbitMQ, CentOS

- Developed during internship at Virtusa for the Colt ASM project.
- Focused on automating incident creation via email parsing and mailbox integration.
- Used by enterprise clients including Wiley, Calcy, and British Telecommunications.

Selected Academic Projects

- Neo Bus Booking System: Full-stack system using React, Spring Boot, Android, MongoDB. [GitHub](#)
- CourseWeb Project: Built using React, Spring Boot, Express.js. [GitHub](#)
- Event Manager with Google Calendar Integration: Java Spring Boot, OAuth 2. [GitHub](#)
- Suzuki Care App: Flutter-based car management system. [GitHub](#)
- CEB Mobile App: Java Android app for utility interaction. [GitHub](#)
- Hospital Management System (WPF C#), Junga Android Game, and other semester-based academic projects.



RESEARCH

Development of a FUI Generator Tool Using JavaFX with Deep Learning & Image Processing

Final Year Research Project - Comprehensive Design & Analysis Project (CDAP), 2020

- Developed a tool to automate the Futuristic User Interface (FUI) design process.
- The system accepts hand-drawn wireframes as input and converts them into enhanced 2.5D designs rendered in a 3D viewport.
- Utilized Generative Adversarial Networks (GANs) for model enhancement, along with image processing techniques.
- Implemented using JavaFX for the interface and deep learning for design recognition and transformation.



PUBLICATION

Indunil Ranawake, Ashhar Ahamed, Sathira Guruge

"Generating 2.5D Motion Graphics from 2D Designs."

Presented at the International Conference on Advancements in Computing (ICAC 2020) - Journal of Applied Research.

I do hereby certify that the above-mentioned information is true and accurate to the best of my knowledge

30-06-2026

Ashhar Ahmed M.A